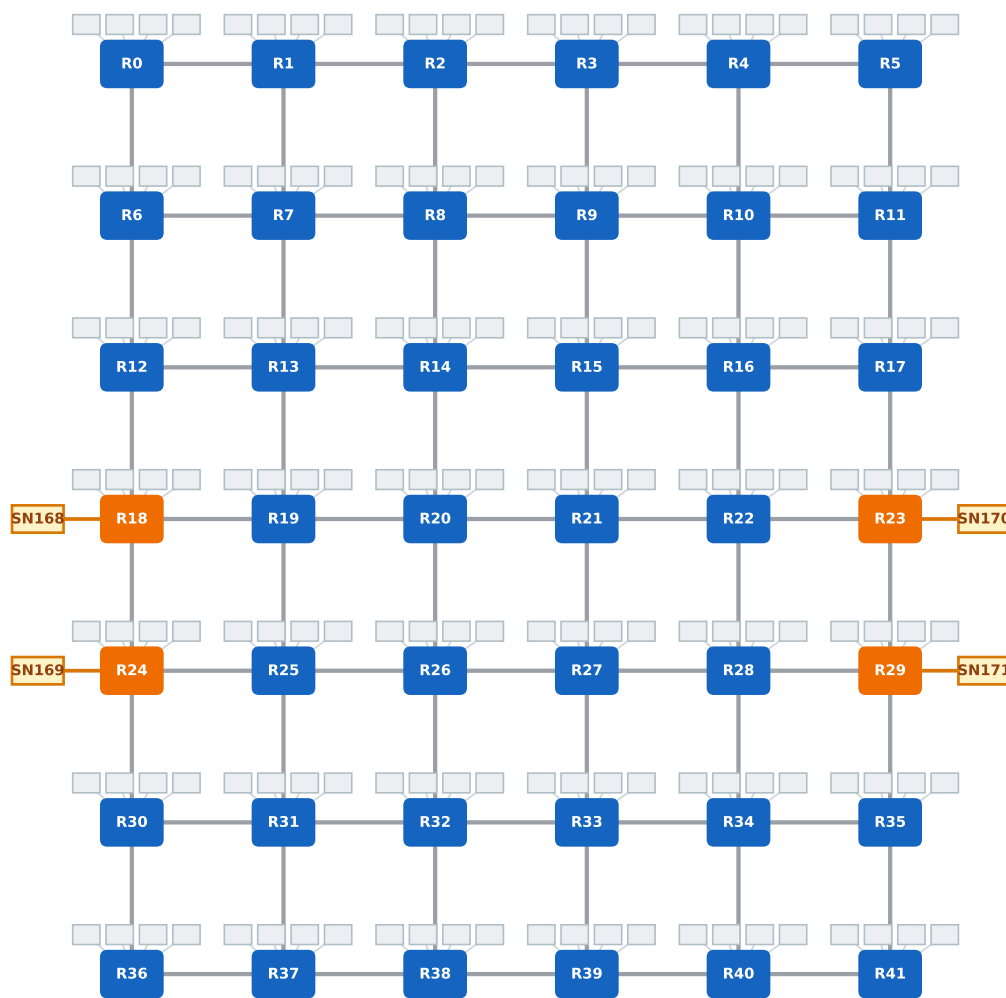
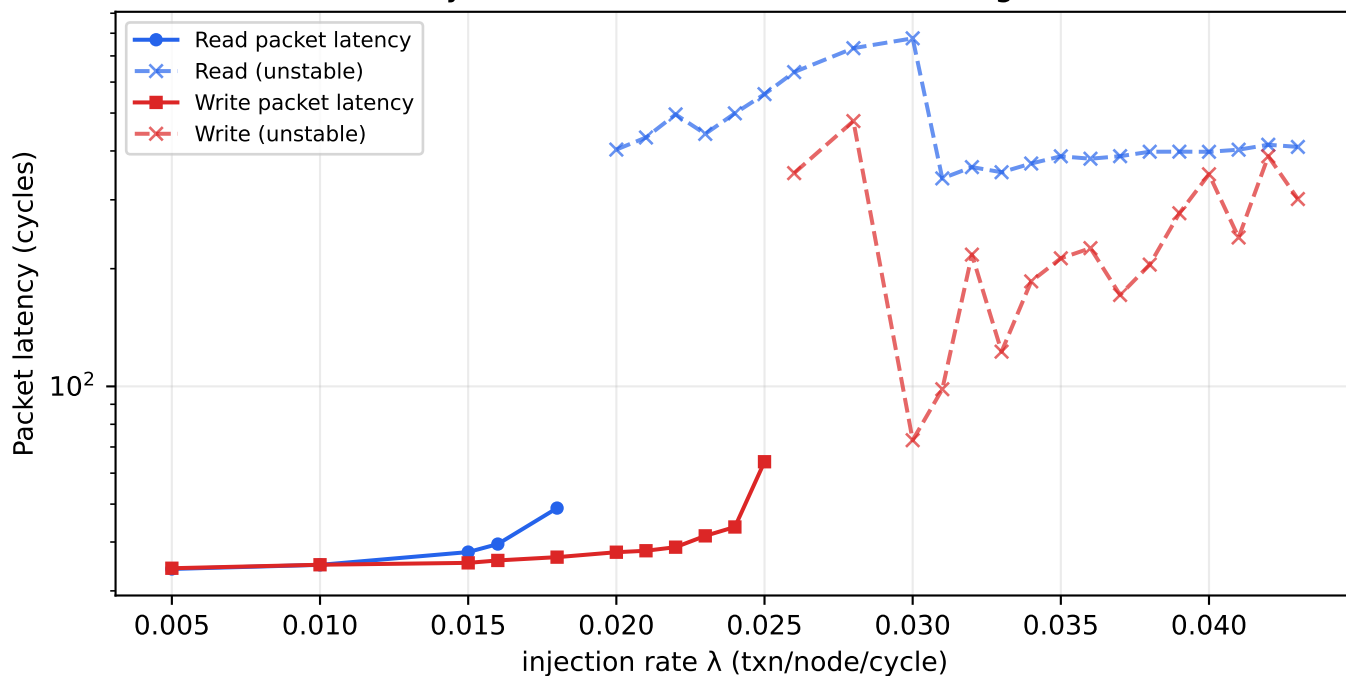


ZCN-LP Simulation Setup

vc buf size=4, DATA FLITS=2, XY, link latency=2
Topology: 7x6 CHI mesh — 84 RN + 84 HN + 4 SN (DDR)
SN @ R18/R24/R23/R29 (nodes 168-171), XY routing, link_latency=2



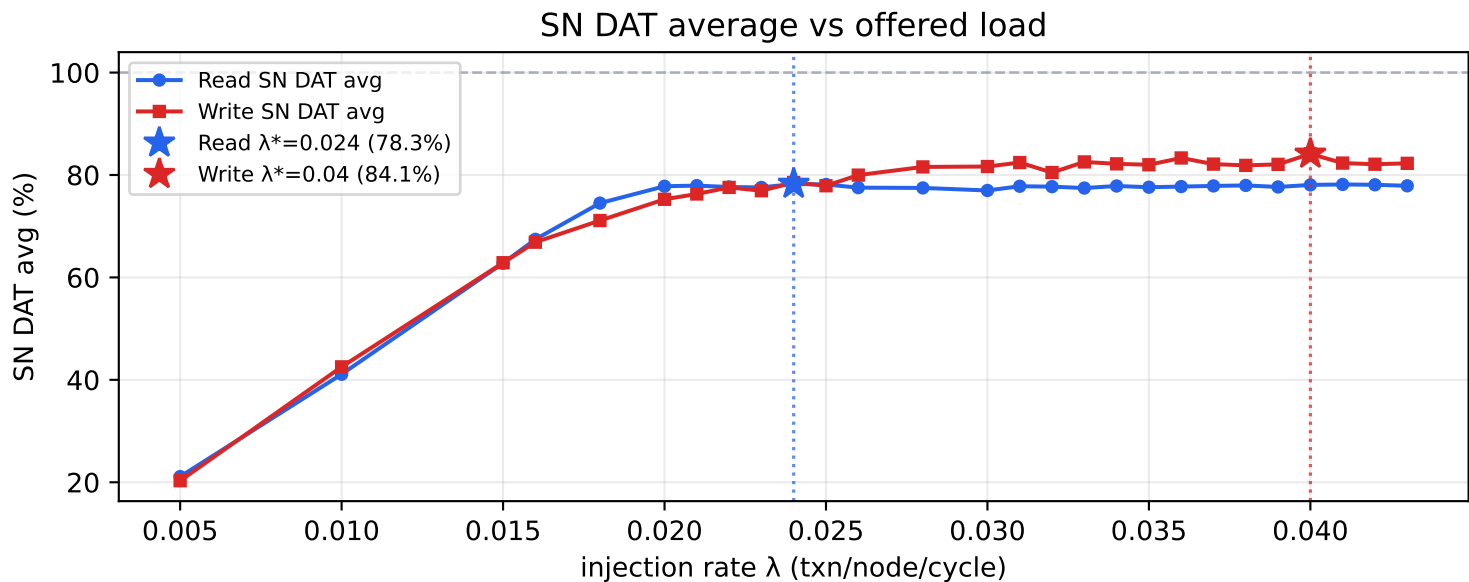
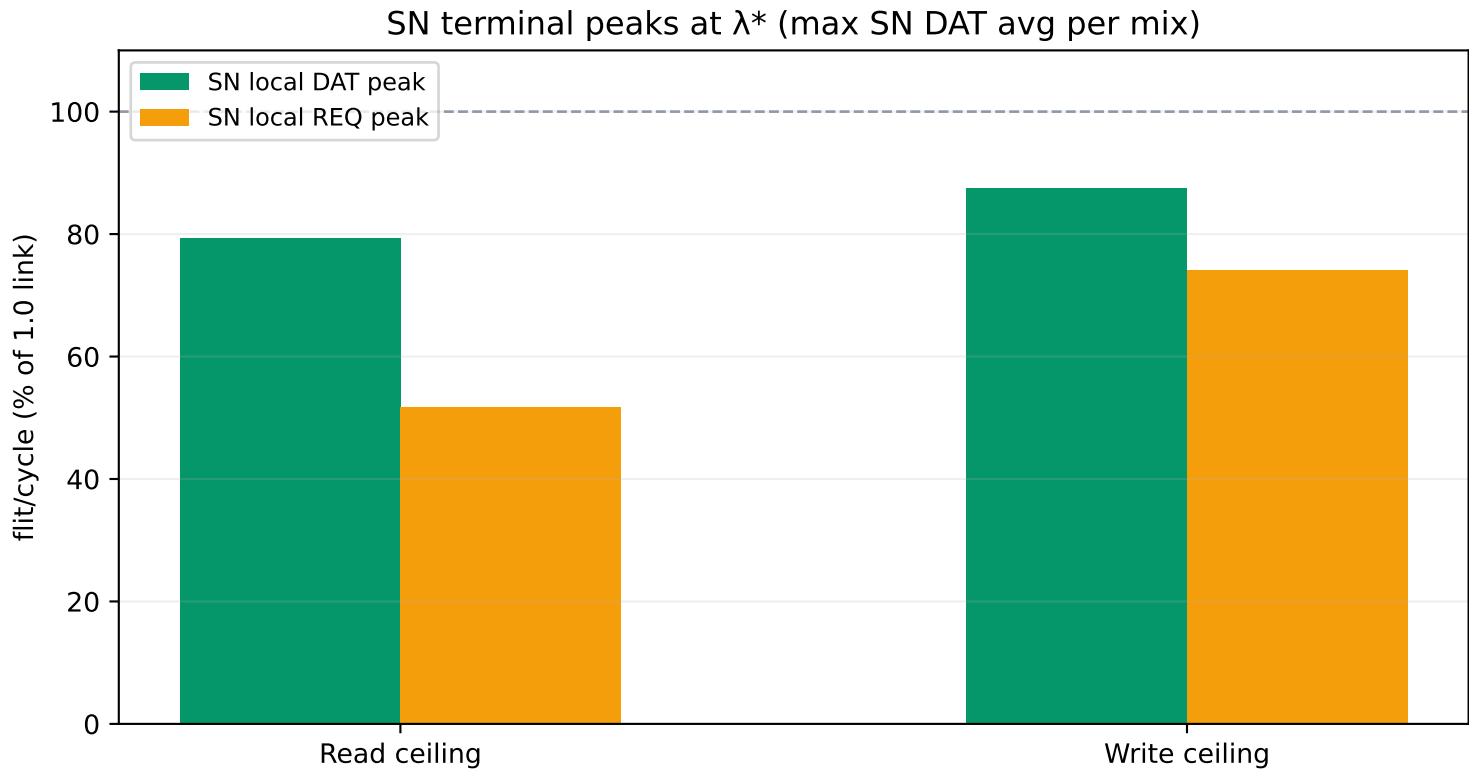
Latency vs offered load (read/write ceiling mixes)



Solid markers = converged (ok). Dashed x = UNSTBL / last DisplayStats snapshot.
 Packet latency (atime-ctime, incl. source queueing) from sweep_sn_throughput ceiling CSVs.

ZCN-LP SN Read/Write Utilisation

vc_buf_size=4 flits/VC, DATA_FLITS=2, XY routing, link_latency=2



Path	λ^* (max avg)	SN DAT peak	SN DAT avg	SN REQ peak	SN REQ avg	State
Read	0.024	79.3%	78.3%	51.7%	49.7%	UNSTBL
Write	0.04	87.5%	84.1%	74.0%	71.2%	UNSTBL

Config: 7×6 mesh CHI, 4 SN nodes @ R18/R24/R23/R29 (168–171), vc_buf=4 flits/VC, 2 VCs, DATA_FLITS=2 (32B/packet on wire).
 Read ceiling: 100% ReadShared DMT miss → CompData SN→RN. Write ceiling: 100% WriteBack + L3EvictToSN.
 SN local DAT read = max sent_flits@SN (inject); write = max accepted_flits@SN (eject). SN local REQ = max accepted_flits@SN on REQ channel to SN.
 Write/read SN DAT avg ratio at λ^* : 1.08× (84% vs 78%).